

KALIAKPA, Bulgaria

WMO: 155620

Lat: 43.3636N

Lon: 28.4657E

Elev: 210

StdP: 14.58

Time Zone: 2.00 (EUE)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.2	20.9	10.8	9.6	16.9	15.0	11.9	22.5	54.5	29.6	53.3	29.0	32.3	0	0.567

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.2	84.1	74.2	82.1	73.1	80.5	72.1	76.6	81.8	74.9	80.3	73.5	79.1	9.8	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.9	131.8	80.8	73.0	123.6	79.4	71.3	116.6	78.0	40.2	82.0	38.6	80.3	37.2	79.2	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 44.0	39.2	27.9	DB	11.2	87.6	6.3	3.3	6.6	90.0	2.9	91.9	-0.6	93.8	-5.2	96.2	(4)
(5)			WB	10.6	78.7	5.7	1.9	6.5	80.0	3.2	81.1	0.0	82.2	-4.1	83.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	55.2	36.7	37.7	42.9	50.3	60.8	69.4	74.5	74.8	67.1	57.9	49.0	40.2	(6)
(7)		DBStd	14.95	8.37	7.72	5.86	4.90	4.91	4.52	3.77	4.15	4.99	6.71	7.25	8.69	(7)
(8)		HDD50	1472	413	344	228	52	1	0	0	0	0	17	103	314	(8)
(9)		HDD65	4442	877	764	686	440	145	14	0	1	37	230	479	769	(9)
(10)		CDD50	3376	1	0	7	62	335	583	760	767	513	263	75	10	(10)
(11)		CDD65	870	0	0	0	0	14	147	295	303	100	11	0	0	(11)
(12)		CDH74	4040	0	0	0	0	3	400	1697	1811	126	3	0	0	(12)
(13)		CDH80	479	0	0	0	0	0	19	207	252	1	0	0	0	(13)
(14)	Wind	WSAvg	13.0	16.0	15.0	14.1	11.9	9.9	9.9	10.8	11.1	12.9	13.9	14.3	16.9	(14)
(15)	Precipitation	PrecAvg	19.0	1.4	0.9	1.4	1.3	1.7	1.7	1.6	1.3	2.5	1.9	1.5	1.6	(15)
(16)		PrecMax	31.7	4.1	2.6	3.9	4.6	3.6	6.8	5.8	3.6	9.6	7.3	3.7	5.4	(16)
(17)		PrecMin	10.6	0.2	0.1	0.1	0.1	0.4	0.2	0.1	0.2	0.2	0.0	0.2	0.2	(17)
(18)		PrecStd	5.0	1.0	0.6	0.8	0.9	0.9	1.3	1.3	1.1	2.0	1.5	0.9	1.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.0	53.7	58.5	66.0	73.8	82.3	86.4	87.3	79.1	74.0	64.8	58.0	(19)
(20)			MCWB	48.8	47.4	49.9	57.0	65.0	72.6	73.8	75.8	70.4	67.8	59.9	54.3	(20)
(21)		2%	DB	50.7	50.3	55.0	61.8	71.3	79.8	83.8	85.1	76.9	70.7	62.7	55.4	(21)
(22)			MCWB	47.2	46.2	48.4	55.2	64.2	70.9	73.5	75.4	68.5	65.7	58.7	52.7	(22)
(23)		5%	DB	48.5	48.3	52.3	58.7	69.4	78.0	82.2	82.6	75.4	68.6	60.3	53.0	(23)
(24)			MCWB	45.6	44.9	47.3	52.8	63.3	70.0	73.1	73.6	67.9	64.2	57.1	50.1	(24)
(25)		10%	DB	46.8	46.6	50.3	56.7	67.6	76.3	80.7	80.8	73.9	66.7	58.3	51.2	(25)
(26)			MCWB	44.4	43.5	46.3	51.8	62.1	69.0	72.2	72.5	67.1	62.4	55.2	48.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.2	48.4	51.2	58.6	67.5	75.0	77.8	81.0	72.8	69.3	61.2	54.8	(27)
(28)			MCDWB	52.9	51.7	56.3	64.0	71.8	80.1	82.7	85.0	76.9	72.3	63.8	57.3	(28)
(29)		2%	WB	47.9	46.8	49.3	55.5	65.7	72.8	76.3	77.7	71.1	66.8	59.1	52.9	(29)
(30)			MCDWB	50.0	49.6	53.4	60.1	69.8	77.9	81.5	83.0	75.1	69.8	62.0	55.3	(30)
(31)		5%	WB	46.0	45.3	47.9	53.8	64.2	71.4	74.8	75.5	69.8	64.7	57.4	50.7	(31)
(32)			MCDWB	48.0	47.8	51.7	57.9	68.3	76.4	80.3	80.8	73.7	68.1	60.2	52.6	(32)
(33)		10%	WB	44.5	43.8	46.7	52.4	62.8	70.0	73.4	73.9	68.4	62.7	55.6	48.9	(33)
(34)			MCDWB	46.6	46.2	50.0	56.0	66.9	75.2	79.3	79.4	72.7	66.1	57.9	51.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.0	7.2	7.4	7.0	7.8	8.6	9.4	9.2	8.5	7.6	7.1	7.4	(35)
(36)			MCDBR	6.6	7.7	9.8	9.5	9.2	10.1	10.3	10.1	9.2	7.9	7.5	7.4	(36)
(37)			MCWBR	5.3	5.4	6.1	5.8	6.7	7.8	8.4	8.5	6.9	6.9	5.6	6.2	(37)
(38)		5% WB	MCDWB	6.2	7.3	8.7	8.3	8.4	9.0	9.1	9.1	8.4	7.7	7.4	7.0	(38)
(39)			MCWBR	5.4	5.6	6.0	5.8	6.7	7.6	8.7	8.6	7.1	7.2	6.1	6.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.356	0.385	0.415	0.457	0.428	0.430	0.438	0.448	0.429	0.398	0.375	0.358	(40)	
(41)		taud	2.333	2.252	2.196	2.093	2.264	2.298	2.286	2.251	2.273	2.329	2.329	2.328	(41)	
(42)		Ebn at Noon	239	251	259	257	269	268	264	257	252	244	229	225	(42)	
(43)		Edh at Noon	27	34	41	49	42	41	41	41	37	31	27	25	(43)	
(44)	All-Sky Solar Radiation	RadAvg	420	658	1072	1547	1968	2157	2154	1926	1371	855	504	377	(44)	
(45)		RadStd	65	80	111	151	109	132	115	90	116	80	66	61	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (19 neighbors)		+0.85	N/A	N/A	N/A	N/A	N/A	N/A	-205	+178	+104

Nomenclature: See separate page